

Automated Resume Ranking Using SpaCy NER and TF-IDF Similarity

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Abstract—Basic keyword matching in resume management systems has given way to advanced AI-powered applications. Although resumes are essential for hiring since they provide a summary of a candidate's qualifications, traditional applicant tracking systems (ATS) frequently fall short due to their limited keyword matching, which ignores contextual relevance. Our AI and Natural Language Processing (NLP) solution integrates machine learning and text categorization algorithms, specifically SpaCy's Named Entity Recognition (NER), to improve resume analysis. This allows for more thorough feature extraction, precisely identifying important components like credentials, experience, and talents. By comparing resumes to job descriptions using a dual-layer scoring approach that includes Cosine Similarity and TF-IDF, the program generates a sorted shortlist of qualified applicants. Using metrics like F1-score, precision, and recall, it offers great accuracy thanks to its built-in Python with scikit-learn.

Index Terms—Resume Screening, Natural Language Processing, Named Entity Recognition, Machine Learning, Candidate Ranking, Recruitment Automation.

I. INTRODUCTION

By improving resume screening systems' intelligence and impartiality, your research fills a significant recruitment vacuum. Both recruiters and candidates regularly lose out on opportunities because traditional application tracking systems (ATS), which frequently rely on simple keyword matching, lack the complexity necessary to evaluate candidates thoroughly. Your method adds a more complex evaluation process by combining machine learning and Natural Language Processing (NLP). A deeper, more thorough understanding of candidate qualifications is made possible by the organized extraction of important data, including skills, education, certifications, and work experience, made possible by the use of SpaCy's Named Entity Recognition (NER) and classification models. In addition to introducing ranking and scoring elements to create a carefully curated shortlist of candidates, this improved analysis better matches resumes with job criteria.

By analyzing CV and job description material objectively, ranking and scoring methods like Cosine Similarity and TF-IDF can be integrated into applicant tracking systems to improve the accuracy of applicant selection. Recruiters may make better hiring decisions by using a feedback loop to continuously learn, and candidates can receive guidance on how to improve the alignment of their resumes. This system, which was created in Python with the aid of modules like SpaCy, scikit-learn, pandas, and NumPy, covers the data processing requirements of resume analysis. Because of its modular design, it may be used alone or integrated into a current applicant tracking system, making it versatile for a variety of recruitment scenarios. In addition to HR, job portals can objectively match candidates to positions and resume optimization helps job seekers create a more fair hiring process.

II. LITERATURE SURVEY

[1] A lot of applicant tracking systems (ATS) rely on basic keyword matching, which is devoid of the advanced expertise required for a more comprehensive analysis of a candidate's credentials. Often, this approach yields biased and constrained screening results.

[2] NER approaches, such as those in SpaCy, enhance the accuracy of candidate profiling and appropriateness assessment by allowing systems to identify key components in resumes, such as firms, roles, and talents.

[3] To enhance candidate-job matching on recruiting platforms, TF-IDF in combination with Cosine Similarity offers a framework for assessing the textual similarity between resumes and job descriptions.

[4] [5] Through the analysis of job-relevant terms across applications, machine learning makes objective evaluation possible and results in precise, automated resume classification and scoring.

[6] Feedback-mechanism systems improve recommendations and better match hiring requirements for individual companies by learning from past hiring decisions.

[7] By making it possible to extract specific candidate traits like experience and qualifications, NLP approaches enhance candidate-job matching.

[8] By reducing human prejudice in resume screening, AI-driven systems that evaluate qualifications objectively contribute to a more egalitarian recruiting process.

[9] [10] [11] Older parsing methods that have trouble understanding complicated resume structures can be outperformed by modern AI-based techniques that better capture candidate details.

[12] [13] Because of its strong NLP capabilities, especially in tokenization and entity recognition, SpaCy is frequently used to process resumes and job descriptions.

[14][15] Recruiters can swiftly find the best applicants and eliminate the need for laborious manual screening by using classification algorithms to rank resumes according to relevance.

[16] [17] Dynamic applicant recommendations that complement job postings are made possible by integrating resume analysis technologies with real-time HR services, such job boards.

[18] By facilitating effective data management and deployment, Python libraries such as scikit-learn aid machine learning applications in hiring.

[19] Automated scoring systems emphasize applicants with the highest scores and expedite the hiring process by using structured data to match resumes to job specifications.

[20] By leveraging structured data to match resumes to job descriptions, automated scoring systems expedite the hiring process by identifying the most qualified applicants based on scores.

III. METHODOLOGY

A. Overview

By automating resume screening, the Resume Analysis for Job Recruitment system uses AI and NLP to improve hiring efficiency and equity. Recruiters must frequently hand analyze resumes as part of traditional hiring, which takes time and may introduce prejudice. The capacity of many

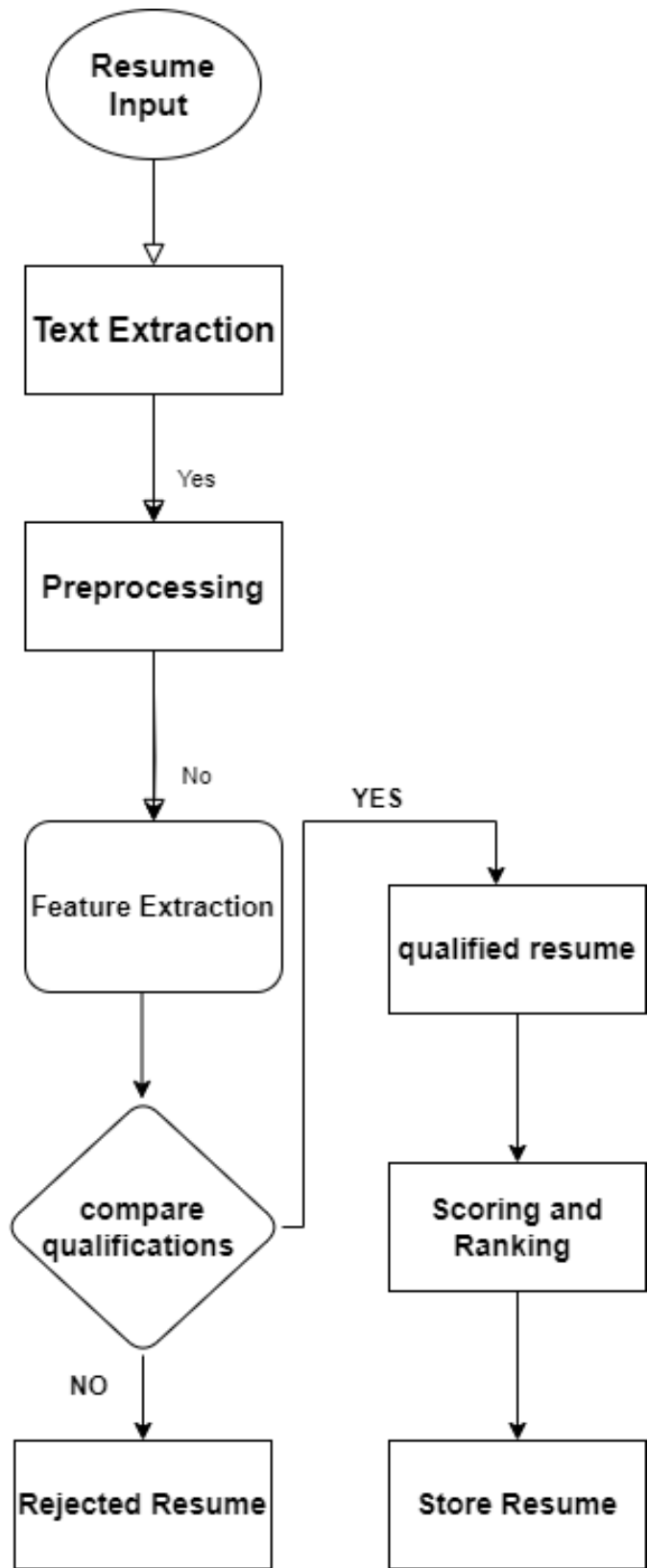


Fig. 1. FLOW DIAGRAM OF RESUME ANALYSIS FOR RECRUITMENT

existing systems, such as Applicant Tracking Systems (ATS), to completely evaluate a candidate's fit is limited because they mostly rely on keyword matching. To assess resumes more precisely, this approach, however, blends machine learning and natural language processing (NLP) with technologies such as SpaCy's Named Entity Recognition (NER). It offers a more comprehensive assessment of application suitability by highlighting important qualifications and assigning scores to applicants according to how well they match job requirements.

B. How the System Works

To identify entities like skills, job titles, and experience, the system first gathers resumes and job descriptions before using SpaCy's NER and tokenization. By evaluating how these elements match the job requirements, it generates candidate profiles. Resumes are then ranked based on relevance scores determined by machine learning models utilizing TF-IDF and Cosine Similarity. This gives recruiters a more curated list of the best applications, and feedback helps hopefuls get better. HR personnel can screen and handle applications in real-time with this technology, which is accessible through a web-based portal.

C. System Workflow

The procedure starts with the resume's text being input and extracted, then preprocessing and feature extraction, as seen in Figure 1. Next, the extracted qualifications are contrasted with the job specifications. The resume is assessed, ranked, and stored if it satisfies the requirements; if not, it is discarded.

D. Gathering Information and Processing Input

The resume rating system starts with the collection and preparation of high-quality data. This entails compiling resumes from job sites, email submissions, and recruitment platforms in a variety of formats, including DOCX, PDF, and plain text. For this project, the majority of resumes were gathered in PDF format.

E. The Ingestion of Data

Using programs like PyMuPDF (sometimes called Fitz), which extracts text from PDFs, resumes are transformed into a machine-readable format after they are gathered. Converting PDFs to text, eliminating formatting and special characters, and saving the processed text in structured formats like CSV or JSON are all examples of data intake. Tokenization, lowercasing, stop word removal, punctuation removal, and lemmatization utilizing SpaCy's NLP package are all examples of preprocessing that guarantees data consistency.

F. Named Entity Recognition (NER)

NER is applied to processed text to identify key entities—skills, experience, and education. For instance, “Python” is tagged as a skill, job titles and company names are categorized under experience, and degree names and institutions are labeled under education. These tagged entities improve the system's feature extraction accuracy and relevance assessment.

G. Feature Extraction

Key resume elements that best reflect the applicant's qualifications are extracted after preprocessing. Word embeddings (e.g., Word2Vec, GloVe) or techniques like TF-IDF are used to transform education, work experience, and skills into vector representations. Effective candidate ranking is supported by the usage of these vectors for resume and job description similarity matching.

H. Data Storage and security

MongoDB, a NoSQL database selected for its scalability, flexibility, and effective querying of semi-structured data, is used to store processed resume data. Documents containing resume ID, applicant name, skills, experience, education, and text are stored by MongoDB. Cloud storage systems like Google Cloud Storage store raw data (PDF, DOCX), providing security features like backup, redundancy, and encrypted storage.

I. Data Retrieval

In order to query and rank resumes according to job descriptions, efficient data retrieval is essential. The following MongoDB retrieval techniques are available through the system:

- **MongoDB Queries:** Resumes can be filtered and sorted using MongoDB's query language according to criteria like education, experience, and talents.
- **Full-Text Search:** The text indexes in MongoDB facilitate keyword or phrase-specific searches and enable full-text search on resume content.
- **Aggregation Framework:** This framework enables sophisticated searches, such as classifying resumes according to experience or skill level and determining ranking scores.

J. Summary

The data storage system in this project leverages MongoDB as a NoSQL database, which provides flexibility, scalability, and efficient querying for resume data. Cloud-based file storage is used for storing raw resume files. The system ensures data security through encryption and access controls, and provides robust data retrieval and backup mechanisms.

K. Model Training and Ranking

The resume rating algorithm evaluates candidate-job fit using supervised machine learning techniques. Training with models like Random Forest, which were selected for their capacity to handle feature interactions, comes after data pre-treatment and feature engineering. During model training, features are fine-tuned and hyperparameters are optimized by cross-validation. To guarantee accurate ranking, performance is assessed using F1-score, recall, accuracy, and precision.

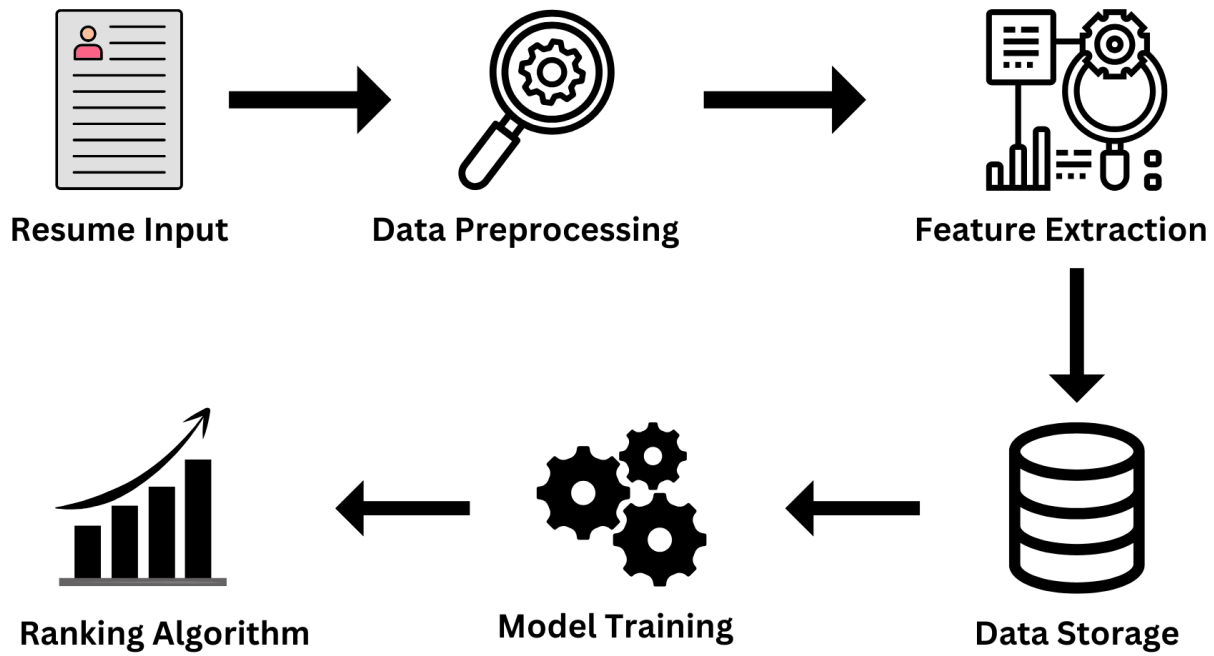


Fig. 2. Architecture Diagram for Resume Analysis

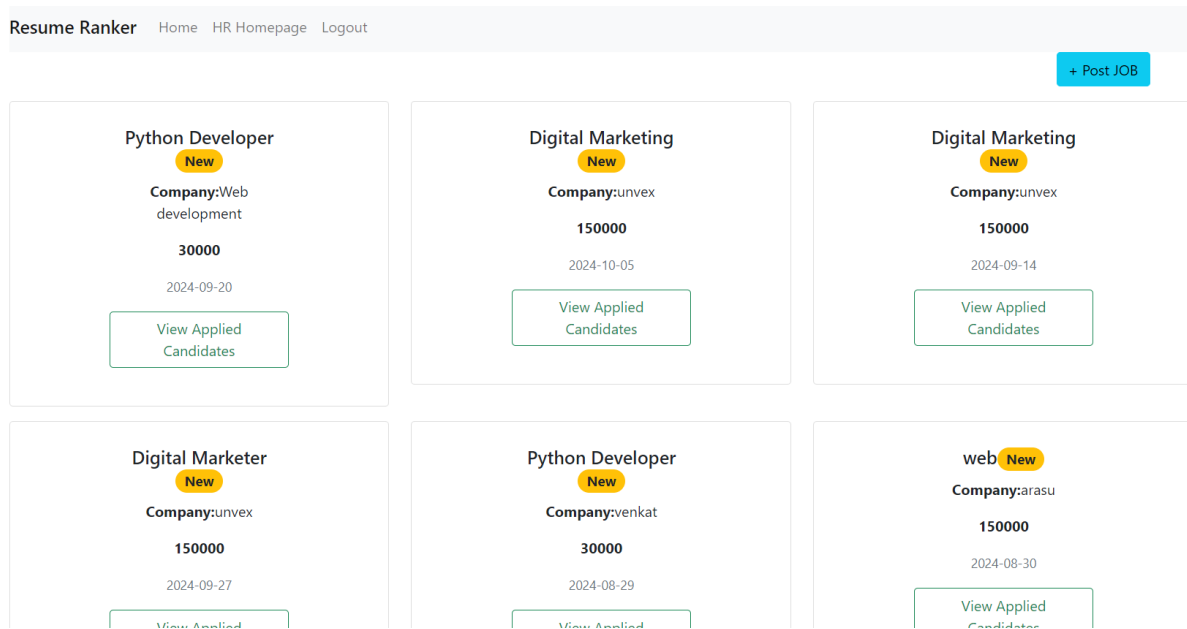


Fig. 3. Job Portal

Applied Candidates



No.	Name	Matching %
1	Arasulingam T	71.67
2	Dineshkumar	68.37
3	Puvirajan	65.23
4	Thilak	62.83
5	Tilak	58.3
6	Ajith	55.89

Fig. 4. Ranked resumes of candidates

L. Model Selection

For resume ranking, a number of machine learning techniques are assessed; a supervised model trained on labeled data is selected. Support Vector Machines (SVM) are excellent at managing high-dimensional data, Random Forest finds non-linear patterns, Neural Networks capture intricate links between resumes and job descriptions, and Logistic Regression offers linear categorization.

M. Model Training

The following are the steps that make up the training process:

- **Dataset Preparation:** A labeled dataset of resumes and corresponding job descriptions is used to train the model. Each resume-job description pair is given a relevance score, which can range from high to low.
- **Feature Engineering:** Features are created by combining job descriptions and resumes. These include keyword matching, experience level, skill overlap, and educational relevance.
- **Model Training:** The chosen model (Random Forest), which takes the extracted features as input, is trained using the labeled data. Hyperparameters such as the maximum depth and number of trees are adjusted by cross-validation.
- **Model Evaluation:** Metrics such as accuracy, precision, recall, and F1-score are used to evaluate the model on a test set to ensure it is operating effectively.

N. Ranking Mechanism

The following ranking process is used after model training:

- **Input Data:** The model is fed a set of resumes from possible applicants as well as the job description.
- **Feature Matching:** The qualities of each CV, including educational background, experience relevance, and talent fit, are compared to the job description.
- **Score Calculation:** The model calculates a relevance score based on each resume's characteristics and how well they match the job description.
- **Ranking:** The resumes are arranged in descending order based on the relevancy scores. The resumes that are most relevant to the job description are the ones that rank highest.

O. Model Deployment

The model is sent into the production environment after it has been trained and adjusted. The model can receive fresh resumes and job descriptions, rank them, and produce the ranked list since it is integrated into the resume ranking system's backend.

P. Evaluation and Metrics

Recall gauges the model's ability to find all relevant resumes, accuracy indicates how well projected ranks match actual ranks, precision evaluates the relevance of top-ranked resumes, and the F1-score strikes a balance between precision and recall. Mean Reciprocal Rank (MRR) assesses the position of the first pertinent resume to offer more information. These

metrics show how well the system works to provide recruiters with a ranked list of the best applicants, improving the hiring process.

Q. Summary

HR users can browse job listings on the Resume Ranker dashboard (Fig. 3). Each job card shows the title, employer, pay, and posting date. Recruiters can view candidates for particular positions by clicking on a "View Applied Candidates" button. HR may easily manage and review applications for a variety of roles by adding new job listings using the "+ Post Job" button in the top-right corner.

IV. CONCLUSION

We created a machine learning-based resume ranking system to automatically classify resumes according to their relevancy to job descriptions. By minimizing manual screening and making sure the best applicants are given priority, we hoped to expedite the recruiting process. Data collection, preprocessing, and feature extraction were the steps in our workflow, and the Random Forest model was chosen since it was good at managing complicated features and producing precise predictions. Resumes are ranked by the algorithm, which calculates a relevancy score based on education, experience, and skills. Its potential in recruiting was demonstrated by evaluation criteria such as accuracy, precision, recall, and F1-score, which demonstrated good precision and recall. Future enhancements can use deep learning, dataset expansion, and user feedback to increase the model's precision and effectiveness.

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